

LINUX

Operating System

Course Title: **Operating System**

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What is Linux?



Open-Source OS

Freely available, modifiable, and distributable by anyone.

Unix-Based Architecture

Built on solid Unix design principles ensuring stability.

Resource Management

Efficiently manages both hardware and software resources.

Free to Use & Modify

No licensing costs — customize to your exact needs.

Key Features of Linux

Multi-User

Multiple users can run sessions simultaneously without conflict.

Multitasking

Runs multiple processes in parallel efficiently.

High Security

Robust permissions and access control system.

Customizable

Modify kernel, UI, and tooling to suit any need.

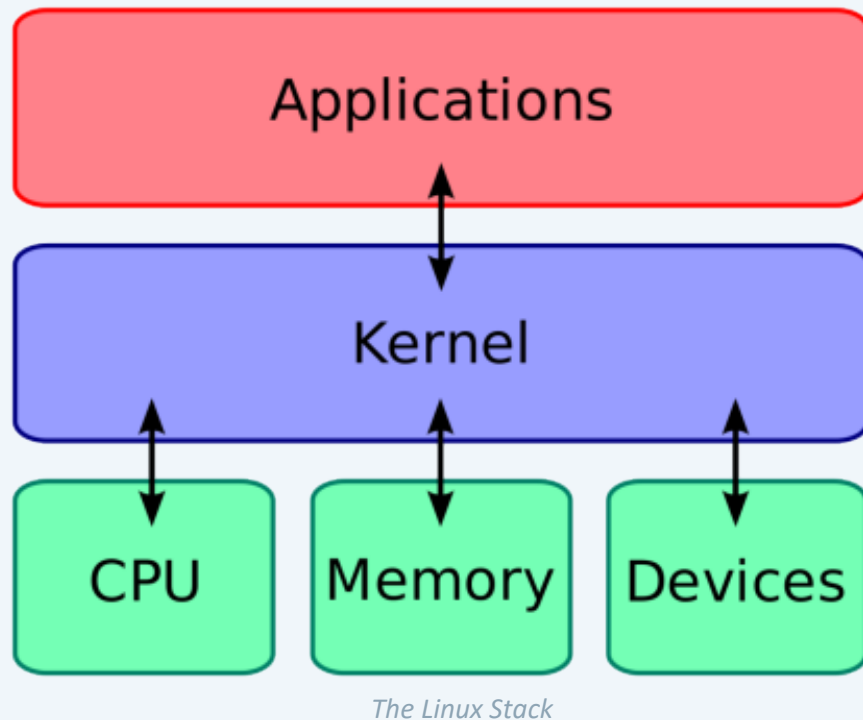
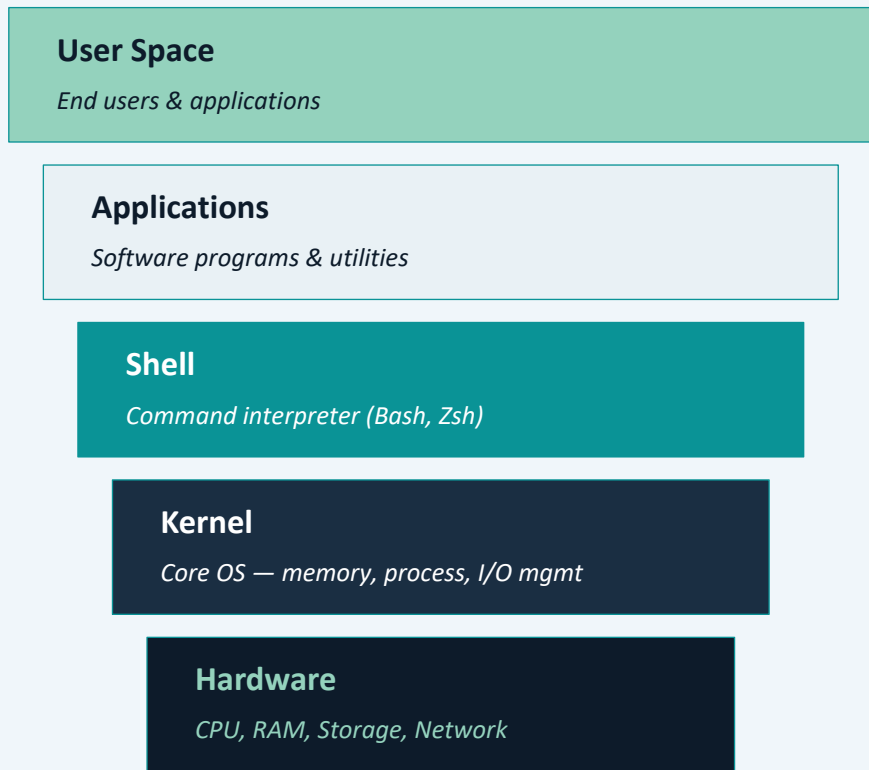
Command Line

Powerful shell interface for complete system control.

Open Source

Community-driven development, no vendor lock-in.

Linux Architecture & Components



Popular Linux Distributions

Ubuntu

Best for beginners & desktop users

Debian

Rock-solid stability & server use

Fedora

Cutting-edge features & developers

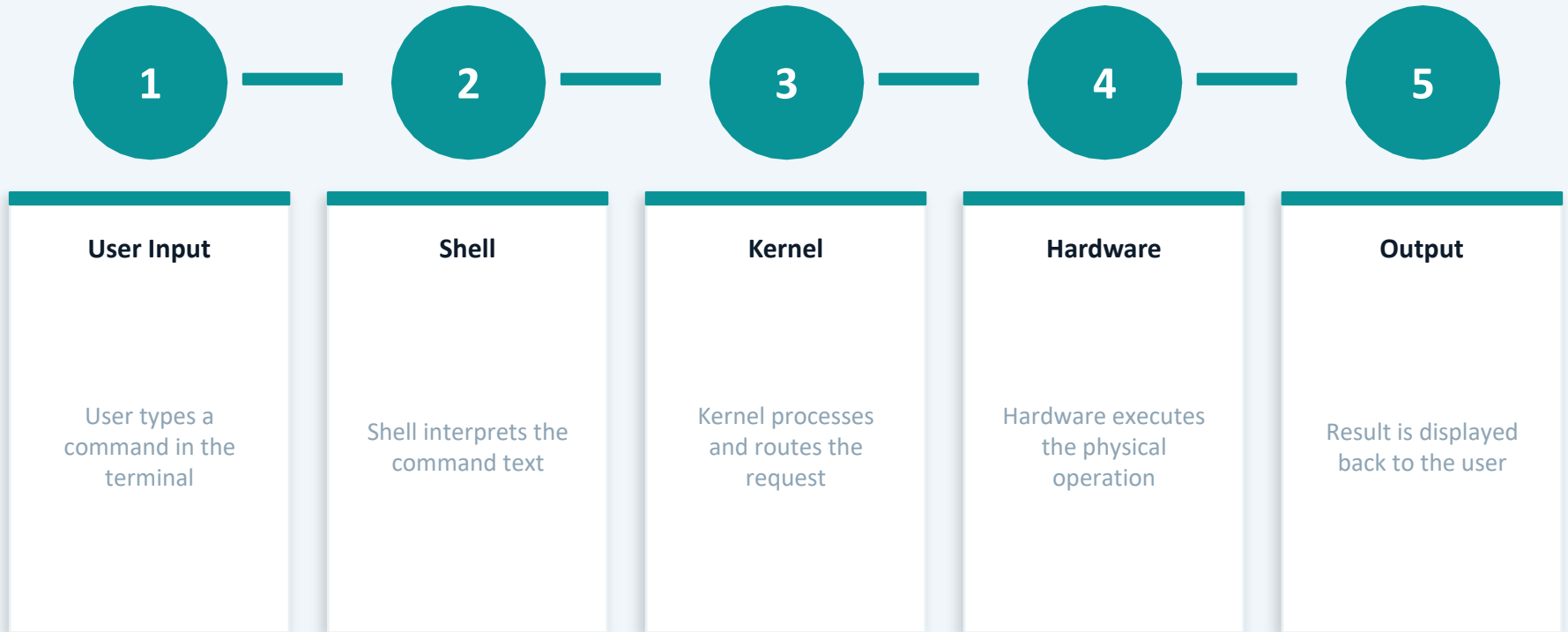
Kali Linux

Cybersecurity & penetration testing

Red Hat

Enterprise-grade commercial Linux

How Linux Works



Advantages vs Disadvantages

✓ Advantages

- ▶ Free and open-source
- ▶ Highly secure and stable
- ▶ Fully customizable
- ▶ Excellent performance
- ▶ Large community support
- ▶ Runs on older hardware

✗ Disadvantages

- ▶ Steeper learning curve
- ▶ Limited software compatibility
- ▶ Requires command-line knowledge
- ▶ Less gaming support
- ▶ Hardware driver gaps

Linux in the Real World

Web Servers

96% of the world's top servers run Linux (Apache, Nginx).

Android OS

Android kernel is based directly on Linux.



Supercomputers

100% of the world's top 500 supercomputers use Linux.

Cloud Infrastructure

AWS, Google Cloud, Azure all run Linux at their core.

Essential Linux Commands

• • • terminal

`$ ls` `# list files and directories`

`$ cd /home` `# change directory`

`$ mkdir projects` `# create a new folder`

`$ rm file.txt` `# delete a file`

`$ pwd` `# print working directory`

`$ sudo apt update` `# update package list`



100 Modern Linux Commands

[CodeForGeek.com](https://codeforgeek.com)

Summary

Linux is a free, open-source operating system with a global developer community.

It is the foundation of web servers, cloud platforms, Android, and supercomputers.

Its layered architecture (hardware → kernel → shell → applications) ensures stability.

Dozens of distributions serve every use case — from desktop to enterprise security.

Thank You

Questions & Discussion Welcome

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